9 **(a)** A stationary wave is formed by the superposition of two progressive waves of the same type, amplitude, frequency and speed, travelling along the same path but in opposite directions.

(b) **(i)** **1.** $\text{Amplitude} = \frac{7.0}{2} = 3.5 \text{ cm}$

2. $\frac{T}{2} = 9.0 \times 10^{-3} \text{ s} \Rightarrow T = 18.0 \times 10^{-3} \text{ s}$

$f = \frac{1}{T} = \frac{1}{18.0 \times 10^{-3}} = 55.56 = 56 \text{ Hz}$

3. $v = \frac{\lambda}{T} = \frac{0.240}{18.0 \times 10^{-3}}$

$= 13.33 = 13.3 \text{ m s}^{-1}$

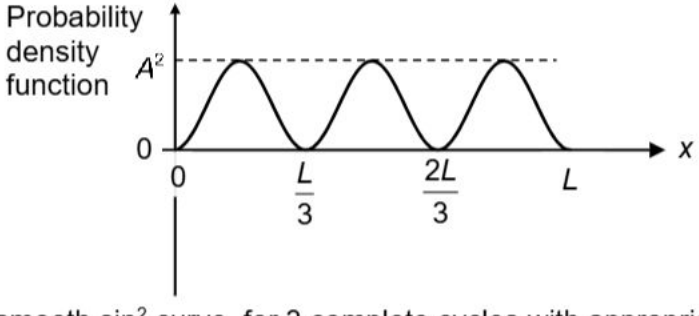
(ii) **1.** 0 rad

2. 3.14 rad (accept π rad)

(c) **(i)** $\int_0^L |\psi|^2 dx$ is the probability of finding the particle between $x = 0$ and $x = L$ of the box. As the box is perfectly reflecting, and the particle is trapped within the box, the probability of finding the particle within the box has to be 1. Hence $\int_0^L |\psi|^2 dx = 1$.

(ii) $\lambda_n = \frac{2L}{n}$

(iii)



*smooth $\sin^2$ curve for 3 complete cycles with appropriate labels.

(iv)

$$\int_0^L |\psi_n(x)|^2 dx = \int_0^L A^2 \sin^2 \frac{2\pi x}{\lambda_n}\, dx$$

$$= \frac{A^2}{2}\int_0^L 2\sin^2 \frac{2\pi x}{\lambda_n}\, dx$$

$$= \frac{A^2}{2}\int_0^L (1-\cos\frac{4\pi x}{\lambda_n})\, dx$$

$$= \frac{A^2}{2}\left[x - \frac{\lambda_n}{4\pi}\sin(\frac{4\pi x}{\lambda_n})\right]_0^L$$

$$= \frac{A^2}{2}\left[x - \frac{2L}{4n\pi}\sin(\frac{4n\pi x}{2L})\right]_0^L$$

$$= \frac{A^2}{2}[L+0] = 1$$

$$\therefore A = \left(\frac{2}{L}\right)^{\frac{1}{2}}$$

OR

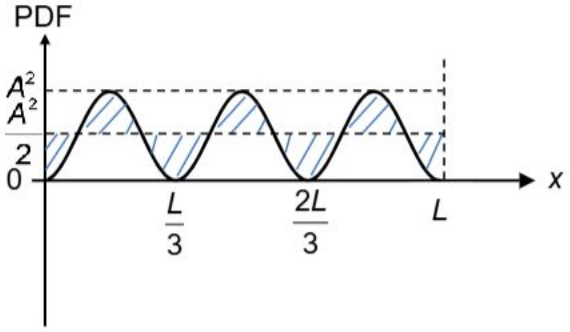
Using graphical method

From (c)(iii)

Since $\int_0^L |\psi_n(x)|^2 dx = 1$

$$\frac{A^2}{2} \times L = 1$$

$$A = \sqrt{\frac{2}{L}}$$



(v) If the walls were not perfectly reflecting, it means that the wave function ψ is non-zero at the walls ($x = 0$, $x = L$). Thus, the particle can penetrate the potential barrier walls and the wave function will decay exponentially within the wall (potential barrier).

Hence there is a finite probability that the particle can be found beyond the boundaries of the box.

(d) (i) The electrons arrive as single (discrete) particles as shown clearly in the first picture.
OR
The dots show that the electrons arrive as single (discrete) particles.

At longer exposure (20 min, 51 s, 5 min 42 s), the regular maxima and minima characteristic of an interference pattern emerges, showing that they exhibit wave-like behaviours.

(ii) Regular fringe separation implies constant wavelength.

According to the de Broglie's postulate, the wavelength of a particle is related to its momentum by $\lambda = \frac{h}{p}$ where h is the Planck's constant. Since wavelength is constant, the momentum of the electrons is constant.

which in turn implies that the electrons have the same velocity.